

U-SPACE study

(SEC) SINGAPORE-ETH
CENTRE

Introduction for Researcher

Case number

Interviewer name

Date of interview (DD/MM/YYYY)

Vision Snellen test

Which is the smallest line that participant can read correctly covering the right eye?

- ☐ 1-7
- ☐ 8
- ☐ 9
- ☐ 10
- ☐ 11

Which is the smallest line that participant can read correctly covering the left eye?

- ☐ 1-7
- ☐ 8
- ☐ 9
- ☐ 10
- ☐ 11

Interview questionnaire

You will be asked several questions during this survey. You do not have to respond to any questions that you feel uncomfortable answering. Please rest assured that all of your responses will be kept strictly confidential and will not be shown to or shared with anyone not connected to this study.

What year were you born in?

What is your gender?

- ☐ Male
- ☐ Female

What gender do you identify with?

- ☐ Male
- ☐ Female
- ☐ Non-binary
- ☐ Other

In which city did you grow up in?

What is your highest level of education?

- ☐ No formal education
- ☐ High school
- ☐ University degree

If university degree, what degree did you graduate with or are currently completing?

What is your occupation?

Do you have a chronic health condition(s)?

(Examples: heart disease, stroke, cancer, hypertension, diabetes, asthma, epilepsy, hypercholesterolaemia)

- ☐ Yes
- ☐ No

Please list your health conditions:

Have you ever suffered a traumatic brain injury?

- ☐ Yes
- ☐ No

How depressed have you felt in the last 6 months?

Rate from 1 (not depressed) to 10 (very depressed).



How anxious have you felt in the last 6 months?

Rate from 1 (not anxious) to 10 (very anxious).



How stressed have you felt in the last 6 months?

Rate from 1 (not stressed) to 10 (very stressed).



Are you a smoker?

- ☐ Yes
- ☐ No

How many cigarettes do you smoke per week?

Do you drink alcohol?

- ☐ Yes
- ☐ No

How many alcoholic drinks do you consume per week?

(One drink is considered to be: half glass of wine, 1 shot of spirit, 1 can of beer)

Have you fallen in the last 12 months?

- ☐ Yes
- ☐ No

How many times have you fallen in the last 12 months?

How many hours do you sleep per day?

How many hours do you walk per week?

How many hours of vigorous exercise do you engage in per week?

Examples: running, swimming, cycling, aerobics or any exercise that make you breathe much harder than normal.

Are you right handed or left handed?

Make sure they list their preferred hand

- ☐ Right handed
- ☐ Left handed

Do you have any experience interacting with a tablet (e.g., surfing the internet)?

- ☐ None
- ☐ Low
- ☐ High

Do you have any previous navigation training (e.g., sea navigation, air navigation, or orienteering)?

- ☐ Yes
- ☐ No

If yes, what type of navigation training have you completed?

Do you have any visual defects?

Examples: astigmatism, myopia, hyperopia

- ☐ Yes
- ☐ No

Please list your visual defects:

Do you usually wear glasses / contact lenses?

- ☐ Yes
- ☐ No

Are you currently wearing your glasses / contact lenses?

Experimenter tells participants to put on vision aid before testing

- ☐ Yes
- ☐ No

What is the strength of your vision aid for your right eye?

What is the strength of your vision aid for your left eye?

Santa Barbara Sense of Direction Scale

This questionnaire consists of several statements about your spatial and navigational abilities, preferences, and experiences. After each statement, you should choose a number on a scale of 1 to 7 to indicate your level of agreement with the statement. Choose "1" if you strongly agree that the statement applies to you, "7" if you strongly disagree, or some number in between if your agreement is intermediate. Choose "4" if you neither agree nor disagree.

	1. Strongly agree	2. Agree	3. Somewhat agree	4. Neither agree nor disagree	5. Somewhat disagree	6. Disagree	7. Strongly disagree
I am very good at giving directions.	☐	☐	☐	☐	☐	☐	☐
I have a poor memory for where I left things.	☐	☐	☐	☐	☐	☐	☐
I am very good at judging distances.	☐	☐	☐	☐	☐	☐	☐
My "sense of direction" is very good.	☐	☐	☐	☐	☐	☐	☐
I tend to think of my environment in terms of cardinal directions (North, South, East, West).	☐	☐	☐	☐	☐	☐	☐
I very easily get lost in a new city.	☐	☐	☐	☐	☐	☐	☐
I enjoy reading maps.	☐	☐	☐	☐	☐	☐	☐
I have trouble understanding directions.	☐	☐	☐	☐	☐	☐	☐
I am very good at reading maps.	☐	☐	☐	☐	☐	☐	☐

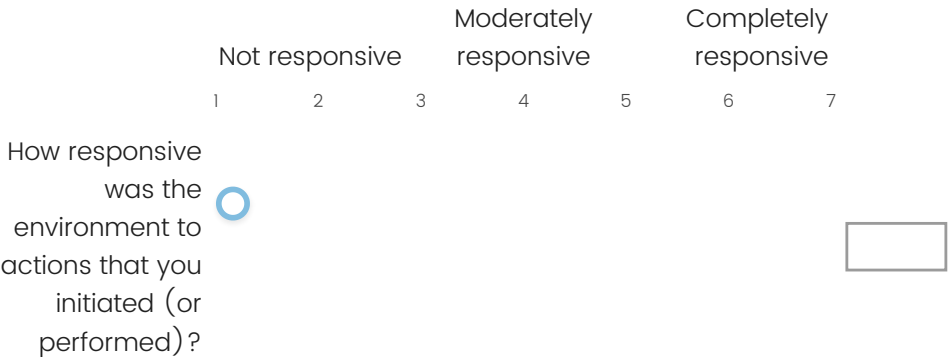
	1. Strongly agree	2. Agree	3. Somewhat agree	4. Neither agree nor disagree	5. Somewhat disagree	6. Disagree	7. Strongly disagree
I don't remember routes very well while riding as a passenger in a car.	☐	☐	☐	☐	☐	☐	☐
I don't enjoy giving directions.	☐	☐	☐	☐	☐	☐	☐
It's not important to me to know where I am.	☐	☐	☐	☐	☐	☐	☐
I usually let someone else do the navigational planning for long trips.	☐	☐	☐	☐	☐	☐	☐
I can usually remember a new route after I have traveled it only once.	☐	☐	☐	☐	☐	☐	☐
I don't have a very good "mental map" of my environment.	☐	☐	☐	☐	☐	☐	☐

SPACE

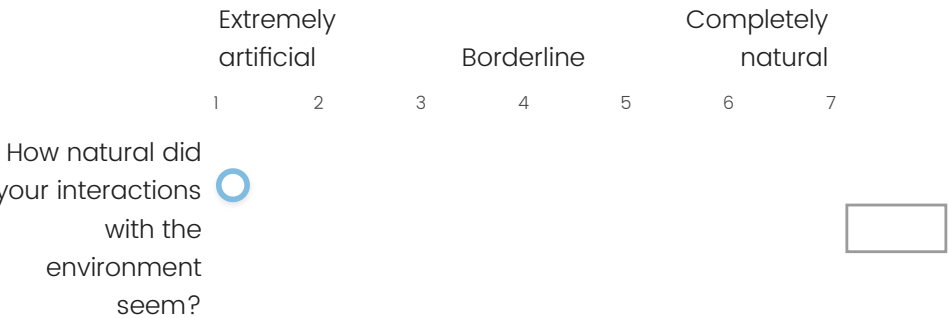


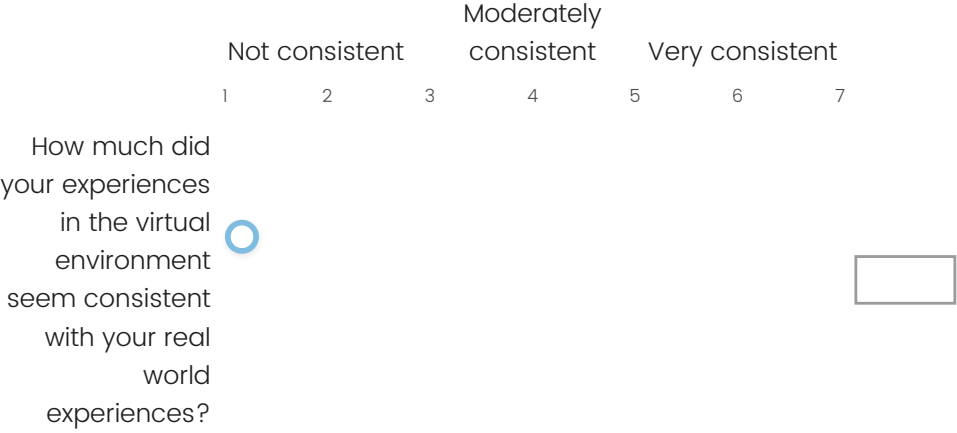
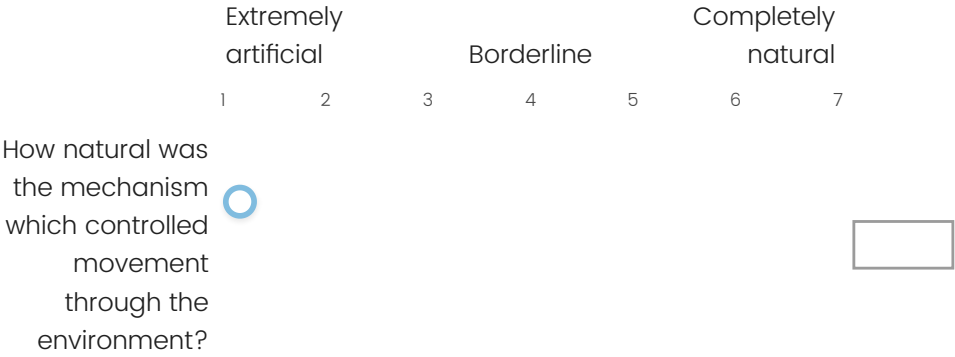
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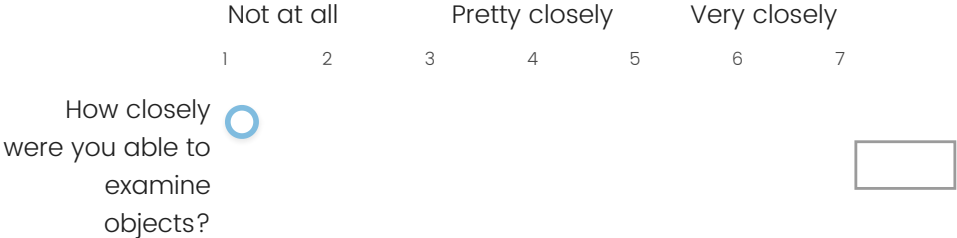
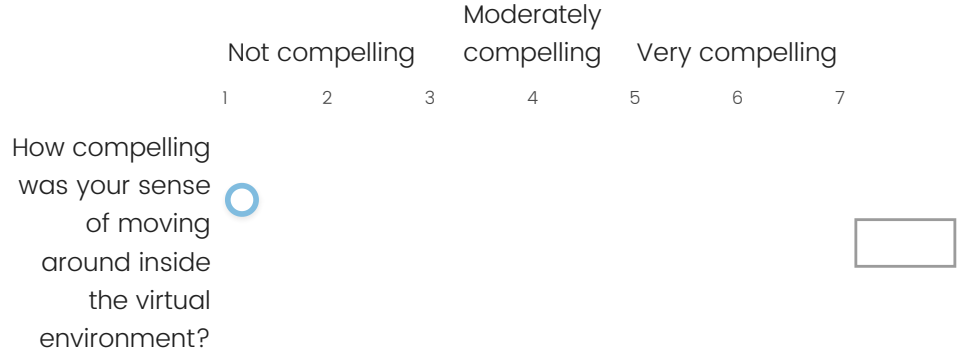


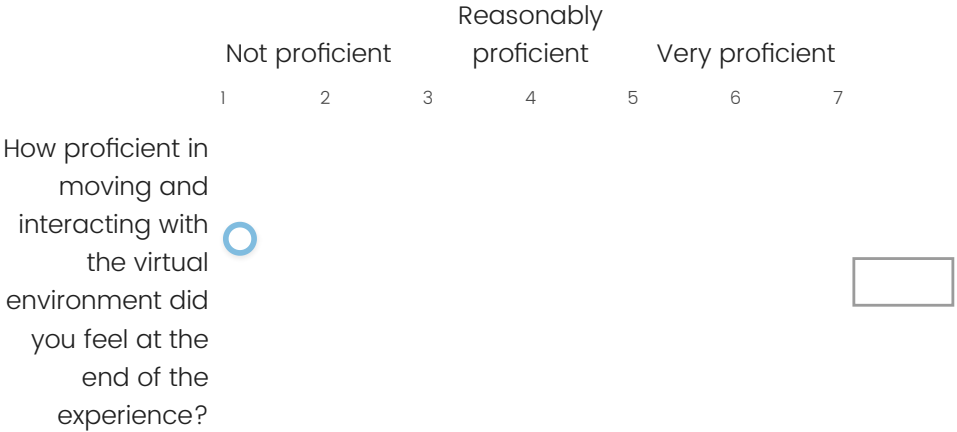
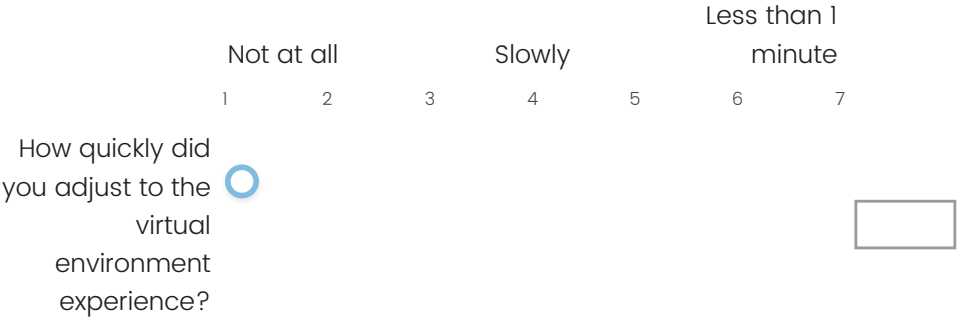
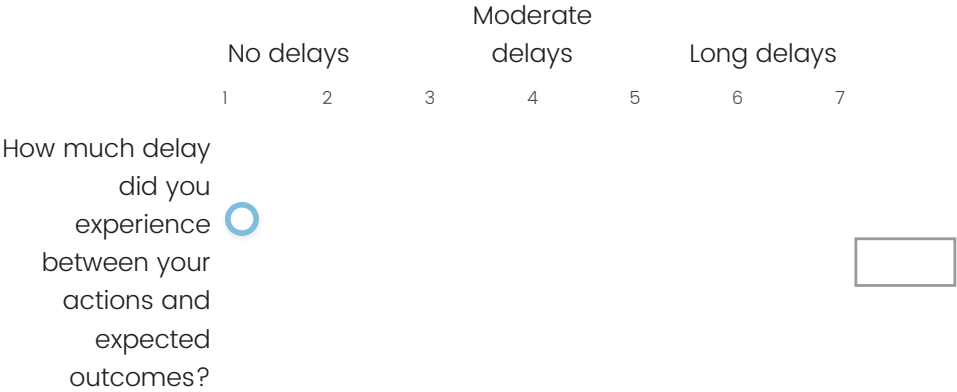


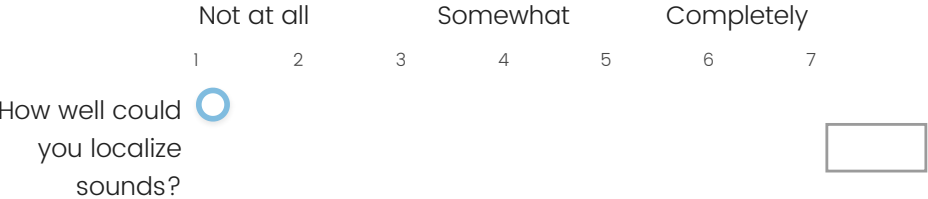
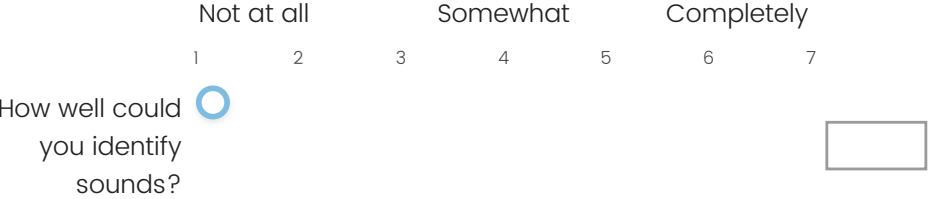
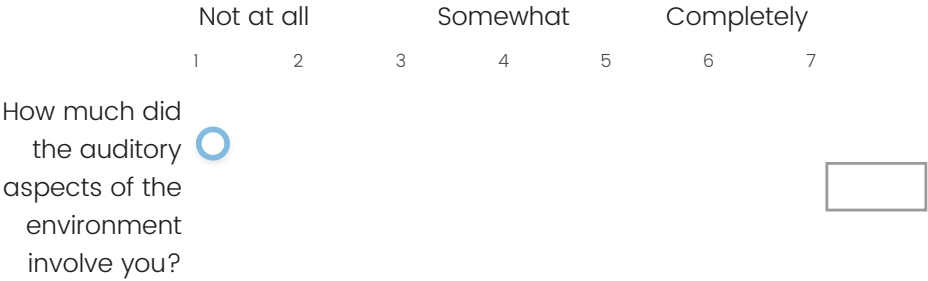
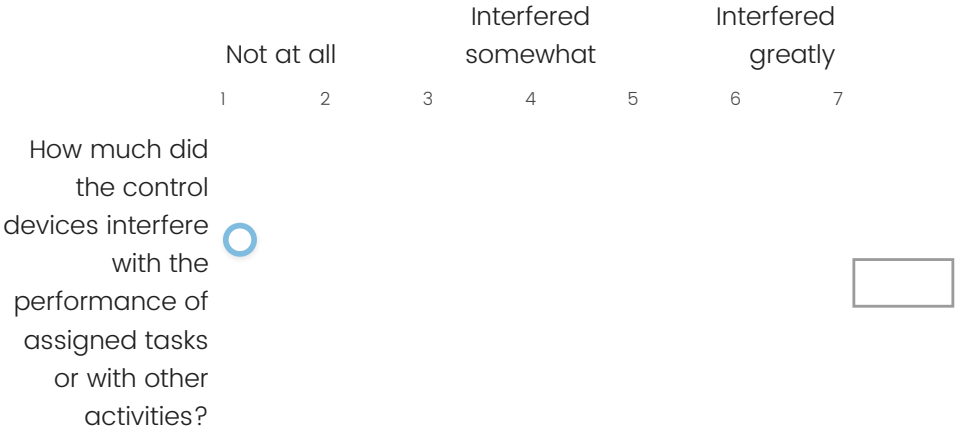
Presence











Usability

For each of the following statements, mark one box that best describes your reactions to the tablet game today.

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I think that I would like to use this tablet game frequently.	☐	☐	☐	☐	☐
I found this tablet game unnecessarily complex.	☐	☐	☐	☐	☐
I thought the game was easy to play.	☐	☐	☐	☐	☐
I think that I would need assistance to be able to use this tablet game.	☐	☐	☐	☐	☐
I found the various functions in this tablet game were well integrated.	☐	☐	☐	☐	☐
I thought there was too much inconsistency in this tablet game.	☐	☐	☐	☐	☐
I would imagine that most people would learn to use this tablet game very quickly.	☐	☐	☐	☐	☐
I found this tablet game very cumbersome/awkward to use.	☐	☐	☐	☐	☐
I felt very confident using this tablet game.	☐	☐	☐	☐	☐

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I needed to learn a lot of things before I could get going with this tablet game.	☐	☐	☐	☐	☐

Please assess the game by ticking one circle per line.

	1	2	3	4	5	6	7	
Annoying	☐	☐	☐	☐	☐	☐	☐	Enjoyable
Not understandable	☐	☐	☐	☐	☐	☐	☐	Understandable
Creative	☐	☐	☐	☐	☐	☐	☐	Dull
Easy to learn	☐	☐	☐	☐	☐	☐	☐	Difficult to learn
Valuable	☐	☐	☐	☐	☐	☐	☐	Inferior
Boring	☐	☐	☐	☐	☐	☐	☐	Exciting
Not interesting	☐	☐	☐	☐	☐	☐	☐	Interesting
Unpredictable	☐	☐	☐	☐	☐	☐	☐	Predictable
Fast	☐	☐	☐	☐	☐	☐	☐	Slow
Inventive	☐	☐	☐	☐	☐	☐	☐	Conventional
Obstructive	☐	☐	☐	☐	☐	☐	☐	Supportive
Good	☐	☐	☐	☐	☐	☐	☐	Bad
Complicated	☐	☐	☐	☐	☐	☐	☐	Easy
Unlikable	☐	☐	☐	☐	☐	☐	☐	Pleasing
Usual	☐	☐	☐	☐	☐	☐	☐	Leading edge

Unpleasant	☐	☐	☐	☐	☐	☐	☐	Pleasant
Secure	☐	☐	☐	☐	☐	☐	☐	Not secure
Motivating	☐	☐	☐	☐	☐	☐	☐	Demotivating
Meets expectations	☐	☐	☐	☐	☐	☐	☐	Does not meet expectations
Inefficient	☐	☐	☐	☐	☐	☐	☐	Efficient
Clear	☐	☐	☐	☐	☐	☐	☐	Confusing
Impractical	☐	☐	☐	☐	☐	☐	☐	Practical
Organised	☐	☐	☐	☐	☐	☐	☐	Cluttered
Attractive	☐	☐	☐	☐	☐	☐	☐	Unattractive
Friendly	☐	☐	☐	☐	☐	☐	☐	Unfriendly
Conservative	☐	☐	☐	☐	☐	☐	☐	Innovative

Feedback_liked

Please tell which aspects of each task you **liked**. Please try to be as detailed as possible. Your comments are very important for improving the game.

Training	
Path integration	
Egocentric pointing	

Mapping

Associative memory

Perspective taking

Please tell which aspects of each task you **disliked**. Please try to be as detailed as possible. Your comments are very important for improving the game.

Training

Path integration

Egocentric pointing

Mapping

Associative memory

Perspective taking

Please write any additional comments you have about SPACE. Your comments are important for improving the game.

Greetings

**(SEC) SINGAPORE-ETH
CENTRE**

Thank you for dedicating time to answer this survey.

**(FHT) FUTURE
HEALTH
TECHNOLOGIES**

Final_information

Instructions

- ☐ Introduced by the researcher
- ☐ Read individually by the participant
- ☐ Video

Control interface

- ☐ Joystick
- ☐ Tap & Swipe
- ☐ Anchor

Game sound

- ☐ Muted
- ☐ Audio

Training time (min) required before starting the navigation assessment

Did the participant passed the training phase?

- ☐ Yes
- ☐ No

Did the participant complete Virtual Reality-Based Spatial Navigation Test?

- ☐ Yes
- ☐ No

Record the hand that participant used for playing the game

- ☐ Left
- ☐ Right
- ☐ Both hands

Did the participant require constant supervision by the researcher?

- ☐ Yes
- ☐ No